

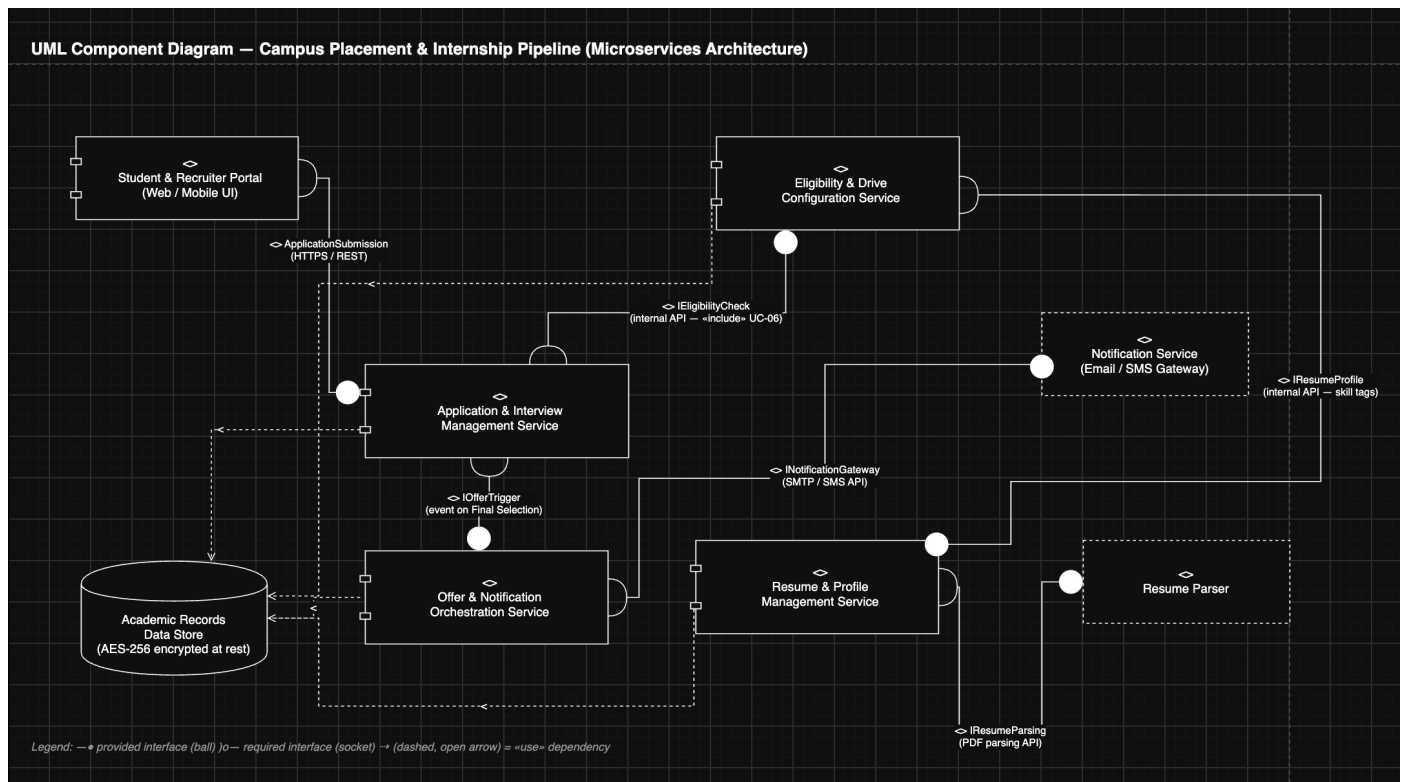
SOFTWARE ENGINEERING

Lab 3: Component Modelling & Architectural Pattern Selection

Problem Statement #03 – Campus & Academic Operations Campus Placement & Internship Pipeline

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1. Component Diagram



2. Architectural Justification

Architecture Selection

We chose Microservices Architecture for the Campus Placement & Internship Pipeline System.

Architectural Choice

The pipeline is decomposed into independently deployable services aligned with the system's core capabilities — resume and profile management, eligibility and drive-rule configuration, application and interview orchestration, and offer/notification handling — each communicating through well-defined interfaces (see accompanying component diagram) rather than through a single monolithic codebase or a single centralized server.

Reason 1 — Independent Scaling Under Peak Load (NFR-002)

Placement drives create short, high-traffic windows — for example, the final hour before an application deadline — where the Application & Interview Management Service and the Eligibility & Drive Configuration Service must sustain 2,000+ concurrent users at a 95th-percentile response time under 3 seconds. Because these are separate services in a microservices architecture, only the hot-path components need to be scaled out horizontally during a drive deadline, while lower-traffic services such as Offer & Notification Orchestration remain at baseline capacity. A layered or client-server design would force the entire application tier (or the single server) to be scaled together, wasting resources and making the 3-second SLA harder to guarantee.

Reason 2 — Independently Evolving Business Capabilities

The placement pipeline serves distinct stakeholders with different release cadences: Placement Officers configuring drives and eligibility rules, Company Recruiters advancing interview rounds, and Student Applicants uploading resumes and tracking status. Each of these maps cleanly to a separate service (Eligibility & Drive Configuration, Application & Interview Management, Resume & Profile Management) that can be developed, tested, and deployed independently — e.g., a change to resume-parsing logic does not require redeploying the interview-scheduling or offer-generation services. This modularity also isolates faults: if the Resume Parser integration degrades, students can still track existing applications and recruiters can still advance interviews.

Security Advantage

NFR-001 requires that all academic records and offer documents be encrypted at rest using AES-256. Isolating persistent data behind a single Academic Records Data Store component, accessed only through a controlled internal interface (IDataAccess) rather than direct storage access from every service, narrows the attack surface: encryption keys and access policies are managed in one place, each service is granted only the minimal data access it needs, and a compromise of one service (e.g., the public-facing Portal) does not automatically expose the underlying encrypted records.

Performance Benefit

Decoupling the Application & Interview Management Service from Offer & Notification Orchestration and Resume & Profile Management means the busiest workflow — students applying and being screened for eligibility during a deadline surge — is not blocked by slower, less time-critical operations such as PDF resume parsing or offer-email dispatch. These can be handled asynchronously (e.g., via events/queues) behind their own service boundaries, keeping the application-submission path fast and helping the system meet its 95th-percentile ≤ 3 -second response-time target even under 2,000 concurrent users.